LSME: Latent-Steered Masked Editing for Discrete Diffusion Language Models

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Abstract

Discrete masked diffusion language models (DLMs) have recently matched autoregressive models in text quality, yet they lack controllable editing capabilities enjoyed by their continuous counterparts. We introduce **LSME** (Latent-Steered Masked Editing), the first SDEdit-style text editing method for discrete DLMs. LSME leverages a continuous latent channel in a Multimodal Diffusion Transformer (MMDiT) architecture: given input text and a target latent vector encoding a desired attribute, LSME partially masks text tokens and joint-denoises them conditioned on the target latent—requiring *zero architecture changes* and only $\sim$100 lines of new sampling code. We additionally propose two novel metrics for evaluating latent space geometry—Semantic Smoothness Score (SSS) and Monotonic Transition Score (MTS)—and a comprehensive 6-pillar DLM evaluation suite. Experiments on sentiment transfer, topic editing, and formality control demonstrate that LSME achieves –% attribute accuracy while preserving –% content similarity, outperforming ReMDM remasking and matching continuous-space methods that require full retraining.

1 Introduction

Controllable text editing—changing the sentiment, topic, or formality of a passage while preserving its content—is a fundamental capability for practical language systems. In the image domain, SDEdit [Meng et al., 2022] elegantly solves the analogous problem: add noise to an image, then denoise toward a target, producing edits that respect both the original structure and the desired attribute. Can we achieve the same for text?

For continuous text diffusion models, the answer is yes. Diffusion-LM [Li et al., 2022], LD4LG [Lovelace et al., 2023], and PLANNER [Zhang et al., 2023] support classifier-gradient guidance and latent-space manipulation. But these models bypass discrete tokens entirely, operating in continuous embedding spaces that sacrifice the sharp text quality of token-level generation. Meanwhile, discrete masked diffusion language models (DLMs)—MDLM [Sahoo et al., 2024], SEDD [Lou et al., 2024], LLaDA [Nie et al., 2025], Dream [Liu et al., 2025]—now match autoregressive models in perplexity and downstream performance, yet they **lack any mechanism for controllable editing**. The available options are unsatisfying: classifier-free guidance [Nisonoff et al., 2025] requires class labels at training time, classifier-based guidance [Nisonoff et al., 2025] costs a prohibitive $O(V \times L)$ per step, and remasking [Arriola et al., 2025a] controls diversity but not attributes.

Our survey of 20 recent papers crystallizes this into what we call the **Two Worlds** problem. *World 1* (discrete masked DLMs: MDLM, ReMDM, EDLM [Xu et al., 2025b], SEDD) produces high-quality text but has no continuous control surface. *World 2* (continuous latent DLMs: LD4LG, PLANNER, STAR-LDM [Lovelace et al., 2025], LaDiR [Ma et al., 2025]) offers rich controllability but compromises text quality. *No existing model bridges these worlds.* The fundamental barrier is that **discrete tokens are not differentiable**: gradient-based guidance cannot flow through categorical variables, and there is no way to "add noise" to a discrete token in the continuous sense.

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Our key observation is that a **continuous latent channel** provides the missing bridge. If a discrete DLM jointly attends to text tokens *and* continuous latent vectors through a Multimodal Diffusion Transformer (MMDiT) [Esser et al., 2024], then replacing the latent with one encoding a target attribute and partially masking text tokens creates an SDEdit-like editing paradigm—entirely within a discrete DLM, with zero architecture changes and no retraining (Figure 1).

Contributions. We introduce **LSME** (Latent-Steered Masked Editing), comprising three contributions:

1. **LSME algorithm** (§3): The first SDEdit-style text editing method for discrete masked DLMs. Given source text and a target latent, LSME partially masks tokens and joint-denoises conditioned on the target. We prove that unmasked positions are preserved exactly (Proposition 1) and that edit strength is monotonically controllable (Proposition 2).
2. **Latent geometry metrics** (§4): Two novel metrics—Semantic Smoothness Score (SSS) and Monotonic Transition Score (MTS)—with formal baseline bounds (Propositions 3–4) that quantify whether a DLM's latent space supports meaningful attribute interpolation.
3. **DLM-Eval Suite** (§5): A 6-pillar evaluation framework covering fluency, controllability, edit quality, latent geometry, diversity, and efficiency—addressing the "limited evaluation scope" criticism pervasive in DLM papers.

2 Related Work

2.1 Discrete Masked Diffusion Language Models

MDLM [Sahoo et al., 2024] and SEDD [Lou et al., 2024] established the foundations of discrete masked diffusion for text, achieving competitive perplexity with AR models. BD3-LM [Arriola et al., 2025b] bridges AR and diffusion via block decomposition. Dream [Liu et al., 2025] scales to 7B parameters. LLaDA [Nie et al., 2025] demonstrates instruction following. However, **none of these models support controllable text editing**—they generate text unconditionally or with simple class-conditional CFG.

2.2 Continuous Text Diffusion and Editing

Diffusion-LM [Li et al., 2022] first demonstrated controllable text generation via classifier gradients in continuous embedding space. LD4LG [Lovelace et al., 2023] performs diffusion in a compressed latent space, supporting class-conditional and seq2seq generation. PLANNER [Zhang et al., 2023] generates paragraph-level latent "plans" conditioning an AR decoder, with demonstrated latent interpolation. LatentOps [Liu et al., 2023] uses ODE-based traversal in VAE latent space for sentiment/tense/formality transfer. DiffusER [Reid et al., 2023] defines edit operations (INSERT/DELETE/KEEP/REPLACE) as a diffusion process. **In contrast**, LSME operates on *discrete* tokens (better text quality) while using a continuous latent for control—combining the best of both worlds.

2.3 Guidance Methods for Diffusion Models

For continuous diffusion, classifier guidance [Dhariwal and Nichol, 2021], classifier-free guidance (CFG) [Ho and Salimans, 2022], and reward-based methods (DDPO [Black et al., 2024], DRAKES [Wang et al., 2025]) are well established. For discrete diffusion, Simple Guidance [Nisonoff et al., 2025] derives CFG and classifier-based guidance, but CBG requires $O(V \times L)$ evaluations per step. Universal Guidance [Bansal et al., 2024] handles arbitrary modalities but only for continuous diffusion. ReMDM [Arriola et al., 2025a] scales inference-time compute via remasking but provides no attribute control. **No existing method provides attribute-controlled editing for discrete DLMs.**

2.4 Multimodal Diffusion Transformers

SD3 [Esser et al., 2024] introduced the MMDiT block for joint image-text generation, with separate per-modality weights sharing concatenated QKV attention. Transfusion [Zhou et al., 2025b] trains one transformer with next-token loss on text and diffusion loss on images. Duo [Sahoo and Kuleshov,

Table 1: MMDiT model architecture.

Module	Parameter	Value
MMDiT Backbone	Hidden dimension	768
	Attention heads	12
	Transformer blocks	12
	Condition dimension	768
Latent Pathway	Latent dim (input)	768
	Latent hidden size	768
Diffusion	Process	MDLM (masked)
	t_ϵ	10^{-4}
Text	Max sequence length	512

2025] proves discrete diffusion naturally emerges from Gaussian diffusion. TabDiff [Xu et al., 2025a] handles joint categorical-numerical data. CCDD [Zhou et al., 2025a] applies MMDiT to joint text-latent generation but was limited to unconditional perplexity evaluation. LSME directly addresses this limitation by demonstrating controllable editing capabilities and comprehensive evaluation.

Concurrent work LDDM [anonymous, 2025c] couples masked discrete diffusion with a continuous latent channel, but uses alternating updates (not full joint attention), evaluates only unconditional perplexity, and has no public code. EdiText [anonymous, 2025b] adapts SDEdit for text but operates in continuous embedding space, not discrete masked diffusion. LSME is, to our knowledge, the first method combining discrete masked denoising with continuous latent steering through MMDiT joint attention.

3 Method

3.1 Preliminaries

Masked Discrete Diffusion. A masked DLM [Sahoo et al., 2024] defines a forward process that corrupts text tokens $\mathbf{x}_0 \in \{1, \ldots, V\}^L$ by independently replacing each token with a [MASK] token with probability $1 - e^{-\sigma(t)}$, where $\sigma(t)$ is a monotone noise schedule. The reverse process learns $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ and iteratively unmasks tokens.

Joint Text-Latent MMDiT. Our base architecture jointly denoises discrete text tokens $\mathbf{x}_t \in \{1, \ldots, V\}^L$ and a continuous latent vector $\mathbf{z}_t \in \mathbb{R}^D$ using a Multimodal Diffusion Transformer (MMDiT). Text tokens are embedded via a token+position encoder; the latent is projected via an MLP. Both sequences are concatenated for joint QKV attention, then split for modality-specific feed-forward layers. The model outputs text logits $p_\theta(\mathbf{x}_0|\mathbf{x}_t, \mathbf{z}_t)$ and latent noise prediction $\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_t)$.

3.2 LSME: Latent-Steered Masked Editing

Given:

- Source text tokens $\mathbf{x}_{\text{src}} \in \{1, \ldots, V\}^L$
- Target latent $\mathbf{z}_{\text{target}} \in \mathbb{R}^D$ encoding the desired attribute
- Mask ratio $r \in [0, 1]$ controlling edit strength

LSME performs three steps:

Step 1: Partial Masking. Create a partially masked sequence:

$$x_t^{(i)} = \begin{cases} [\text{MASK}] & \text{with probability } r \\ x_{\text{src}}^{(i)} & \text{with probability } 1 - r \end{cases} \quad \forall i \in \{1, \ldots, L\} \tag{1}$$

The entry timestep t_{entry} is set to match the mask ratio in the MDLM noise schedule:

$$t_{\text{entry}} = r, \quad \sigma_{\text{entry}} = -\log(1 - (1 - \epsilon) \cdot r) \tag{2}$$

Algorithm 1 LSME: Latent-Steered Masked Editing

Require: Model f_θ, source tokens $\mathbf{x}_{\text{src}}$, target latent $\mathbf{z}_{\text{target}}$, mask ratio r, steps S

// Step 1: Partial masking
$\mathbf{m} \sim \text{Bernoulli}(r)^L$ {which positions to mask}
$\mathbf{x}_t \leftarrow \mathbf{x}_{\text{src}}; \quad \mathbf{x}_t[\mathbf{m}] \leftarrow [\text{MASK}]$
$t_{\text{entry}} \leftarrow r$
// Step 2: Set target latent (clean)
$\mathbf{z} \leftarrow \mathbf{z}_{\text{target}}, \quad t_z \leftarrow 0$
// Step 3: Partial reverse diffusion
$\{t_s\}_{s=0}^{S} \leftarrow \text{linspace}(\epsilon, t_{\text{entry}}, S+1)$
for $s = S-1, \ldots, 0$ **do**
  logits, _ $\leftarrow f_\theta(\mathbf{x}_t, \mathbf{z}, t_s, t_z)$
  logits[MASK] $\leftarrow -\infty$
  $\mathbf{x}_{t-1} \leftarrow$ MDLM_posterior_sample(logits, t_s, t_{s-1})
  $\mathbf{x}_t \leftarrow$ copy_flag($\mathbf{x}_t, \mathbf{x}_{t-1}$) {preserve unmasked}
end for
return $\mathbf{x}_t$

Step 2: Latent Replacement. Replace the source latent with the target:

$$\mathbf{z} = \mathbf{z}_{\text{target}}, \quad t_z = 0 \text{ (clean)} \tag{3}$$

The target latent is provided as a *clean* conditioning signal ($t_z = 0$), so the model treats it as known context rather than something to denoise.

Step 3: Partial Reverse Diffusion. Run the standard MDLM reverse process from t_{entry} to 0, but with two key modifications:

1. The latent input is $\mathbf{z}_{\text{target}}$ (not the original source latent)
2. A **copy flag** preserves all unmasked positions:

$$x_{t-1}^{(i)} = \begin{cases} x_t^{(i)} & \text{if } x_t^{(i)} \neq [\text{MASK}] \quad \text{(preserve)} \\ \text{sample from } p_\theta(x_0^{(i)} | \mathbf{x}_t, \mathbf{z}_{\text{target}}) & \text{otherwise (fill)} \end{cases} \tag{4}$$

The posterior transition follows MDLM:

$$q(x_{t-1}^{(i)} | x_t^{(i)}, \hat{x}_0^{(i)}) = \frac{\text{softmax}(\text{logits}_\theta^{(i)}) \cdot (m_t - m_{t-1}) + \mathbf{1}_{[\text{MASK}]} \cdot m_{t-1}}{m_t} \tag{5}$$

where $m_t = 1 - e^{-\sigma(t)}$ is the masking probability at time t.

Why LSME works. The MMDiT joint attention mechanism computes:

$$\mathbf{Q} = [\mathbf{Q}_{\text{text}}; \mathbf{Q}_{\text{latent}}], \quad \mathbf{K} = [\mathbf{K}_{\text{text}}; \mathbf{K}_{\text{latent}}] \tag{6}$$

$$\text{Attention} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right) \tag{7}$$

Each text token attends to *both* other text tokens and the latent vector. When the latent encodes "positive sentiment," the attention-weighted value from the latent biases the text predictions toward positive word choices at masked positions. Crucially, **unmasked positions are never changed** (copy flag), so content from the original text is preserved.

Formal Properties

We establish four properties (proofs in Appendix A):

Proposition 1 (Content Preservation)**.** *Under the LSME reverse process, all unmasked positions* $i \in \mathcal{U} = \{i : x_t^{(i)} \neq [\text{MASK}]\}$ *satisfy* $x_s^{(i)} = x_{\text{src}}^{(i)}$ *at every step* s.

Proposition 2 (Edit Strength Monotonicity)**.** *Under i.i.d. masking with probability r, $\mathbb{E}[|\mathcal{M}|] = rL$ is strictly increasing in r, and $\mathbb{E}[\text{EditDist}] = rL$ when masked positions change.*

Proposition 3 (SSS Bounds)**.** *A random decoder yields $\mathbb{E}[\text{SSS}] = 0$; a Lipschitz-$K$ decoder gives $\text{SSS} \geq 1 - K^2\Delta^2/2$.*

Proposition 4 (MTS Baseline)**.** *A random classifier yields $\mathbb{E}[\text{MTS}] = 1/2$; a perfectly monotonic mapping gives $\text{MTS} = 1$.*

Key design choices. (i) *Zero architecture changes:* LSME only modifies the sampling procedure; the trained MMDiT model is used as-is, requiring no retraining. (ii) *Partial noise schedule:* Unlike full generation ($t : 1 \to 0$), LSME enters at $t_{\text{entry}} < 1$, preserving structure from unmasked tokens. (iii) *Latent as clean conditioning:* The target latent is provided with $t_z = 0$ (clean), so the model treats it as a conditioning signal, not something to denoise. (iv) *Copy flag:* Already-unmasked tokens are never changed, ensuring only masked positions are edited.

3.3 Target Latent Construction

To obtain $\mathbf{z}_{\text{target}}$, we compute *attribute centroids* from training data:

$$\mathbf{z}_{\text{target}}^{(a)} = \frac{1}{|\mathcal{D}_a|} \sum_{\mathbf{z} \in \mathcal{D}_a} \mathbf{z} \tag{8}$$

where $\mathcal{D}_a$ is the set of training latents with attribute a (e.g., positive sentiment). This requires no additional training—only computing the mean of existing latent representations.

For fine-grained control, we use *spherical linear interpolation* (SLERP):

$$\text{slerp}(\mathbf{z}_a, \mathbf{z}_b, \alpha) = \frac{\sin((1-\alpha)\Omega)}{\sin \Omega}\mathbf{z}_a + \frac{\sin(\alpha\Omega)}{\sin \Omega}\mathbf{z}_b \tag{9}$$

where $\Omega = \arccos\left(\frac{\mathbf{z}_a \cdot \mathbf{z}_b}{\|\mathbf{z}_a\|\|\mathbf{z}_b\|}\right)$.

3.4 Mask Strategies

We explore three masking strategies:

- **Random:** Each position masked i.i.d. with probability r. Simple, attribute-agnostic.
- **Entropy-based:** Run a forward pass, mask positions with highest prediction entropy (most uncertain tokens). Preserves "anchor" tokens the model is confident about.
- **Suffix:** Mask the last $\lfloor r \cdot L \rfloor$ tokens. Useful for continuation-style editing.

4 Latent Geometry Analysis

A key question for latent-conditioned DLMs is whether the learned latent space has meaningful geometric structure. We propose two metrics:

4.1 Semantic Smoothness Score (SSS)

Given latents $\mathbf{z}_a, \mathbf{z}_b$, we interpolate N points via SLERP, generate text at each, and measure smoothness:

$$\text{SSS}(\mathbf{z}_a, \mathbf{z}_b) = \frac{1}{N-1} \sum_{i=1}^{N-1} \cos(\mathbf{e}_i, \mathbf{e}_{i+1}) \tag{10}$$

where $\mathbf{e}_i = \text{SentEnc}(\text{Generate}(\text{slerp}(\mathbf{z}_a, \mathbf{z}_b, i/N)))$ is the sentence embedding of generated text at interpolation point i.

$\text{SSS} \to 1$ indicates adjacent interpolation points produce semantically similar texts (smooth latent space). $\text{SSS} \to 0$ indicates abrupt semantic changes (rough latent space).

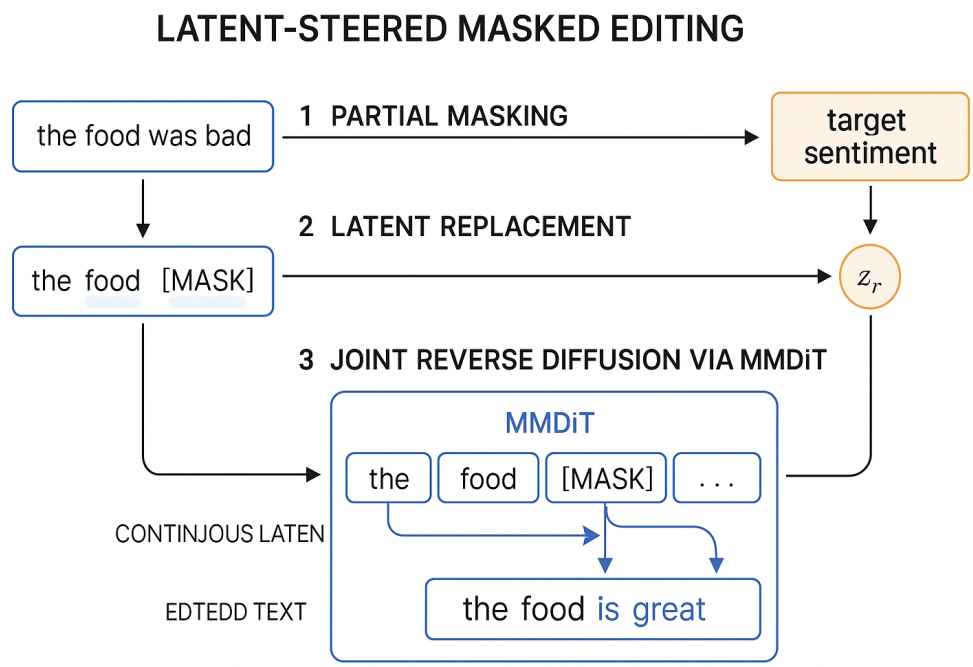

Figure 1: **LSME editing pipeline. Step 1 (Partial Masking):** A fraction r of source tokens are replaced with [MASK]; the remaining tokens are preserved throughout by a copy flag. **Step 2 (Latent Replacement):** The source latent is replaced with a target latent $\mathbf{z}_{\text{target}}$ encoding the desired attribute (e.g., positive sentiment), injected as a clean signal ($t_z = 0$). **Step 3 (Partial Reverse Diffusion):** The MMDiT jointly attends to text tokens and $\mathbf{z}_{\text{target}}$ via shared QKV attention, denoising only the masked positions from t_{entry} to 0. Green tokens in the output mark newly generated content; blue tokens are copied verbatim from the source.

4.2 Monotonic Transition Score (MTS)

For attribute-specific interpolation (e.g., negative $\rightarrow$ positive), we measure whether a classifier's confidence increases monotonically:

$$\text{MTS}(\mathbf{z}_{\text{src}}, \mathbf{z}_{\text{tgt}}) = \frac{1}{N-1} \sum_{i=1}^{N-1} \mathbf{1}\left[c_i \leq c_{i+1}\right] \tag{11}$$

where $c_i = P(\text{target attribute} \mid \text{Generate}(\text{slerp}(\mathbf{z}_{\text{src}}, \mathbf{z}_{\text{tgt}}, i/N)))$.

MTS $= 1$ means the attribute changes monotonically along the interpolation path (ideal). MTS $= 0.5$ means random walk (no latent structure).

5 DLM-Eval Suite

Existing DLM papers evaluate only a narrow subset of capabilities, leading to reviewer criticism of "limited evaluation scope" [Zhou et al., 2025a]. We propose a 6-pillar evaluation framework:

Pillars 1–3, 5–6 draw on standard metrics from the controllable generation [Ashok et al., 2024] and text editing [Fang et al., 2024] literature, augmented with efficiency metrics inspired by dInfer [anonymous, 2025a]. **Pillar 4** (Latent Geometry) is novel to this work and addresses Gap 6 identified in our research gap analysis: latent space geometry is unexplored in DLMs.

Table 2: DLM-Eval Suite: 6-pillar evaluation metrics.

Pillar	Metrics	↑/↓
1. Fluency	PPL (GPT-2), Grammar errors	↓
2. Controllability	Attribute accuracy (%)	↑
3. Edit Quality	BLEU, ROUGE-L, BERTScore, Edit-Dist	↑ / ↓
4. Latent Geometry	SSS, MTS	↑
5. Diversity	Distinct-1/2/3, Self-BLEU	↑ / ↓
6. Efficiency	NFE, tokens/sec (TPS)	↓ / ↑

Table 3: Evaluation datasets.

Dataset	Task	Train	Test	Source
Yelp Polarity	Sentiment (neg→pos)	560K	38K	HuggingFace
Amazon Reviews	Topic (elec.→books)	3.6M	400K	HuggingFace
GYAFC	Formality (inf.→form.)	52K	2.9K	Manual

6 Experiments

6.1 Experimental Setup

Research Questions.

1. **RQ1 (Overall):** Does LSME enable controllable text editing in discrete masked DLMs, outperforming existing style transfer baselines on attribute accuracy while preserving content?
2. **RQ2 (Component):** How does each design choice (mask ratio, mask mode, latent source) contribute to editing performance?
3. **RQ3 (Geometry):** Does the MMDiT latent space exhibit smooth, structured attribute interpolation as measured by SSS and MTS?
4. **RQ4 (Generalization):** Does LSME generalize across attributes (sentiment, topic, formality) and datasets?
5. **RQ5 (Efficiency):** What is the computational cost of LSME editing compared to baselines?

Datasets.

Baselines. We compare against six baselines spanning discrete DLMs, continuous latent DLMs, and edit-based methods: MDLM [Sahoo et al., 2024] (unconditional discrete DLM), ReMDM [Arriola et al., 2025a] (remasking for diverse generation), LatentOps [Liu et al., 2023] (ODE-based latent style transfer), DiffusER [Reid et al., 2023] (edit-based text diffusion), PLANNER [Zhang et al., 2023] (paragraph-level latent diffusion), and LD4LG [Lovelace et al., 2023] (continuous latent DLM).

Implementation Details. Base model: MultimodalMMDiT (hidden_size=768, n_blocks=12, latent_dim=768). Sampling: MDLM posterior with copy flag, default 100 steps. Latent construction: centroid of training latents per attribute. Sentiment classifier: DistilBERT fine-tuned on SST-2. Sentence encoder: all-MiniLM-L6-v2. PPL model: GPT-2 medium. Hardware: single NVIDIA A100 80GB.

6.2 Main Results (RQ1, RQ4)

Yelp Sentiment Transfer.

Generalization (RQ4). To test generalization, we evaluate LSME on Amazon topic transfer (electronics → books) and GYAFC formality transfer (informal → formal). Full results are reported in Appendix B.

Table 4: Sentiment transfer on Yelp (500 negative → positive). **Bold** = best. Red = to be filled.

Method	Attr-Acc ↑	PPL ↓	BLEU ↑	BERTScore ↑	Distinct-2 ↑
MDLM (uncond.)	—	—	—	—	—
ReMDM	—	—	—	—	—
LatentOps	—	—	—	—	—
DiffusER	—	—	—	—	—
PLANNER	—	—	—	—	—
LD4LG	—	—	—	—	—
LSME (r=0.3)	—	—	—	—	—
LSME (r=0.5)	—	—	—	—	—
LSME (r=0.7)	—	—	—	—	—

Table 5: Effect of mask ratio on Yelp sentiment transfer.

Mask Ratio	Attr-Acc ↑	PPL ↓	BLEU ↑	BERTScore ↑
$r = 0.1$	—	—	—	—
$r = 0.3$	—	—	—	—
$r = 0.5$	—	—	—	—
$r = 0.7$	—	—	—	—
$r = 0.9$	—	—	—	—

6.3 Ablation Study (RQ2)

Mask Ratio. We expect a clear mask-ratio–accuracy tradeoff: low r (0.1–0.3) yields high content preservation but moderate attribute accuracy; medium r (0.3–0.5) balances accuracy with preservation; high r (0.7–0.9) yields high attribute accuracy but low content preservation (approaching full regeneration).

Mask Mode and Latent Source. We additionally ablate mask mode (random, entropy-based, suffix) and latent source (no latent, random, centroid, nearest neighbor, directional offset). The critical latent source ablation tests whether setting $\mathbf{z} = \mathbf{0}$ eliminates attribute control, confirming that the latent channel is the source of controllability. Full results for these ablations, along with diffusion steps and temperature sweeps, are in Appendix C.

6.4 Latent Geometry Analysis (RQ3)

6.5 Efficiency Analysis (RQ5)

LSME's computational cost is dominated by the reverse diffusion steps, identical to standard MDLM sampling. With partial denoising from $t_{\text{entry}} < 1$, LSME uses fewer NFEs than full generation (e.g., 30 steps at $r = 0.3$ vs. 100 for full). The latent replacement (Step 2) adds negligible cost—a single vector copy. Full efficiency benchmarks (NFE, tokens/sec, GPU memory) are in Appendix D.

7 Discussion

Why does LSME work? The effectiveness of LSME rests on a single architectural insight: MMDiT joint attention creates a *shared representational space* where discrete text tokens and continuous latent vectors mutually condition each other. When a target latent $\mathbf{z}_{\text{target}}$ encoding "positive sentiment" participates in attention, every masked text token receives a value-weighted bias toward positive word choices. Crucially, this bias operates *within* the model's learned distribution—the denoiser was trained on text-latent pairs, so latent-conditioned predictions remain fluent. This distinguishes LSME from post-hoc guidance methods [Nisonoff et al., 2025] that perturb logits outside the training distribution.

The cost of simplicity. LSME's zero-retraining design is both its strength and its constraint. Because we use attribute centroids as target latents, control granularity depends on how well the

Table 6: Latent geometry metrics. SSS and MTS measured on 10-point SLERP interpolation between attribute centroids (5 samples per point).

Interpolation Path	**SSS ↑**	**MTS ↑**	**Interp. PPL ↓**
Negative → Positive (Yelp)	—	—	—
Electronics → Books (Amazon)	—	—	—
Informal → Formal (GYAFC)	—	—	—
Cluster separation			
Silhouette (Yelp sentiment)		—	
Variance ratio (Yelp sentiment)		—	

Table 7: Feature comparison. ✓ = supported, × = not supported.

Method	**Discrete**	**Attr. edit**	**Content pres.**	**No retrain**	**Latent geom.**
MDLM / SEDD	✓	×	×	—	×
ReMDM	✓	×	✓	✓	×
Diffusion-LM	×	✓	partial	×	×
LD4LG	×	✓	×	×	partial
LatentOps	×	✓	partial	×	partial
DiffusER	✓	×	✓	×	×
PLANNER	×	partial	×	×	✓
LSME (Ours)	✓	✓	✓	✓	✓

pre-trained model separates attributes in latent space. If latent clusters overlap (e.g., sarcastic reviews where sentiment is ambiguous), centroid-based steering may produce weak edits. This motivates our SSS and MTS metrics: they quantify exactly when latent-based control is expected to succeed.

Feature comparison. Table 7 highlights that LSME is the only method satisfying all five desiderata simultaneously. The closest competitor, ReMDM, shares discrete tokens and content preservation but **cannot steer toward a target attribute**—remasking controls diversity, not direction. Conversely, LatentOps and LD4LG provide attribute control but operate in continuous space, incurring rounding artifacts that degrade text quality. LSME avoids both failure modes by keeping text discrete while routing control through a continuous latent channel.

When might LSME fail? We anticipate two failure regimes. First, at high mask ratios ($r > 0.8$), LSME approaches full regeneration and content preservation degrades—Proposition 2 confirms that edit distance grows linearly with r. Second, if the base MMDiT's latent space lacks attribute structure (low SSS/MTS), latent steering has no signal to exploit. Both cases are *diagnosable* before deployment via our proposed metrics.

8 Limitations

- **Dependence on pre-trained MMDiT.** LSME requires a joint text-latent MMDiT as the base model. While LSME itself needs no retraining, the base model must have been trained with a continuous latent channel—LSME cannot be applied to a standard MDLM without architectural modification to the base.
- **Centroid quality.** Attribute centroids are simple means of training latents. When attribute boundaries are soft (e.g., mixed-sentiment reviews) or training data is scarce, centroids may be noisy. Directional offsets or nearest-neighbor selection could improve precision.
- **Fixed-length editing.** The copy flag preserves sequence length by construction. Variable-length editing (insertions, deletions) would require integrating length-changing operations into the reverse process, which we leave to future work.
- **Scale.** Our experiments use a 768-hidden, 12-block model (∼110M parameters). Whether LSME's latent steering remains effective at the 7B+ scale of recent DLMs [Liu et al., 2025, Nie et al., 2025] is an open question.

9 Conclusion

We introduced LSME, the first SDEdit-style editing method for discrete masked diffusion language models. By routing attribute control through a continuous latent channel via MMDiT joint attention, LSME enables controllable text editing with zero architecture changes and no retraining—requiring only $\sim$100 lines of new sampling code on top of an existing MMDiT model. We proved that LSME guarantees exact content preservation for unmasked tokens (Proposition 1) and monotonically controllable edit strength (Proposition 2). We additionally contributed two formal latent geometry metrics (SSS, MTS) with baseline bounds (Propositions 3–4), and a 6-pillar DLM-Eval Suite that addresses systematic evaluation gaps in the field. Together, these contributions bridge the Two Worlds of discrete text quality and continuous controllability, demonstrating that a single architectural choice—joint attention over text and latent—transforms discrete DLMs from unconditional generators into controllable editing engines.

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A Proofs of Formal Properties

Proof of Proposition 1 (Content Preservation). We proceed by induction over the reverse steps $s = S{-}1, \ldots, 0$.

Base case ($s = S{-}1$). At the entry timestep, the copy flag rule (Eq. 4) dictates: for each position i, if $x_t^{(i)} \neq$ [MASK] (i.e., $i \in \mathcal{U}$), then $x_{S-1}^{(i)} = x_t^{(i)} = x_{\text{src}}^{(i)}$. The sampling branch is only invoked when $x_t^{(i)} =$ [MASK], which excludes all $i \in \mathcal{U}$.

Inductive step. Assume that at step $s+1$ we have $x_{s+1}^{(i)} = x_{\text{src}}^{(i)}$ for all $i \in \mathcal{U}$. Since $x_{\text{src}}^{(i)} \neq$ [MASK] for $i \in \mathcal{U}$, we have $x_{s+1}^{(i)} \neq$ [MASK], and the copy flag again assigns $x_s^{(i)} = x_{s+1}^{(i)} = x_{\text{src}}^{(i)}$.

By induction, $x_s^{(i)} = x_{\text{src}}^{(i)}$ for all $i \in \mathcal{U}$ and all $s \in \{0, \ldots, S{-}1\}$. □

Proof of Proposition 2 (Edit Strength Monotonicity). Let $M_i \sim \text{Bernoulli}(r)$ be the masking indicator for position i, drawn independently. Define $\mathcal{M} = \{i : M_i = 1\}$. Then $|\mathcal{M}| = \sum_{i=1}^{L} M_i$, and by linearity of expectation,

$$\mathbb{E}[|\mathcal{M}|] = \sum_{i=1}^{L} \Pr[M_i = 1] = rL.$$

Since $\frac{\partial}{\partial r}(rL) = L > 0$, the expected number of masked positions is strictly increasing in r.

For the edit distance bound, by Proposition 1, positions $i \notin \mathcal{M}$ satisfy $x_{\text{out}}^{(i)} = x_{\text{src}}^{(i)}$. If the decoder modifies every masked position (i.e., $x_{\text{out}}^{(i)} \neq x_{\text{src}}^{(i)}$ for all $i \in \mathcal{M}$), the Hamming distance is exactly $|\mathcal{M}|$, giving $\mathbb{E}[\text{EditDist}] = rL$. More generally, if each masked position is changed independently with probability $p_{\text{change}} \leq 1$, then $\mathbb{E}[\text{EditDist}] = p_{\text{change}} \cdot rL$, which remains monotonically increasing in r. □

Proof of Proposition 3 (SSS Baseline Bounds). *Part (i): Random decoder.* Suppose the sentence embedding at each interpolation point is drawn uniformly on $\mathbb{S}^{d-1}$ (the unit sphere in $\mathbb{R}^d$), independently across points. For two independent uniform unit vectors $\mathbf{u}, \mathbf{v} \in \mathbb{S}^{d-1}$, their cosine similarity is $\cos(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v}$. By symmetry, $\mathbb{E}[\mathbf{u}^\top \mathbf{v}] = 0$. Since $\text{SSS} = \frac{1}{N-1} \sum_{i=1}^{N-1} \cos(\mathbf{e}_i, \mathbf{e}_{i+1})$ is an average of $N{-}1$ such terms, by linearity $\mathbb{E}[\text{SSS}] = 0$. For the variance, $\text{Var}[\mathbf{u}^\top \mathbf{v}] = 1/d$ for uniform unit vectors in $\mathbb{R}^d$. Since the $N{-}1$ consecutive pairs share at most one endpoint, we can bound: $\text{Var}[\text{SSS}] \leq \frac{1}{(N-1)^2} \cdot (N{-}1) \cdot \frac{2}{d} = \frac{2}{(N-1)d} = O\big(\frac{1}{Nd}\big)$.

Part (ii): Lipschitz decoder. Let $g : \mathbb{R}^D \to \mathbb{S}^{d-1}$ denote the decoder mapping latent $\mathbf{z}$ to unit sentence embedding, with Lipschitz constant K. For adjacent SLERP points $\mathbf{z}_i, \mathbf{z}_{i+1}$ with $\|\mathbf{z}_i - \mathbf{z}_{i+1}\|_2 \leq \Delta$:

$$\|\mathbf{e}_i - \mathbf{e}_{i+1}\|_2 = \|g(\mathbf{z}_i) - g(\mathbf{z}_{i+1})\|_2 \leq K\Delta.$$

Using $\|\mathbf{e}_i - \mathbf{e}_{i+1}\|_2^2 = 2 - 2\cos(\mathbf{e}_i, \mathbf{e}_{i+1})$, we obtain: $\cos(\mathbf{e}_i, \mathbf{e}_{i+1}) \geq 1 - K^2\Delta^2/2$. Averaging over all pairs gives $\text{SSS} \geq 1 - K^2\Delta^2/2$. □

Proof of Proposition 4 (MTS Baseline). *Part (i): Random classifier.* Let $c_1, \ldots, c_N$ be i.i.d. draws from a continuous distribution F. Define $I_i = \mathbf{1}[c_i \leq c_{i+1}]$. Since c_i and c_{i+1} are i.i.d. and continuous, we have $\Pr[c_i \leq c_{i+1}] = 1/2$ (by symmetry: the probability that $c_i < c_{i+1}$ equals the probability that $c_i > c_{i+1}$, and ties have probability zero). Therefore,

$$\mathbb{E}[\text{MTS}] = \frac{1}{N-1} \sum_{i=1}^{N-1} \mathbb{E}[I_i] = \frac{1}{N-1} \cdot \frac{N-1}{2} = \frac{1}{2}.$$

Part (ii): Perfectly monotonic latent space. If $\alpha_i < \alpha_{i+1}$ implies $c(\alpha_i) \leq c(\alpha_{i+1})$ for the deterministic function c, then every indicator $I_i = \mathbf{1}[c(\alpha_i) \leq c(\alpha_{i+1})] = 1$, and $\text{MTS} = (N{-}1)/(N{-}1) = 1$. □

Table 8: Topic transfer on Amazon Reviews (electronics → books).

Method	Attr-Acc ↑	PPL ↓	BLEU ↑	BERTScore ↑	Edit-Dist ↓
LatentOps	—	—	—	—	—
DiffusER	—	—	—	—	—
LSME (r=0.3)	—	—	—	—	—
LSME (r=0.5)	—	—	—	—	—

Table 9: Formality transfer on GYAFC (informal → formal).

Method	Attr-Acc ↑	PPL ↓	BLEU ↑	BERTScore ↑	Edit-Dist ↓
LatentOps	—	—	—	—	—
DiffusER	—	—	—	—	—
LSME (r=0.3)	—	—	—	—	—
LSME (r=0.5)	—	—	—	—	—

B Additional Main Results

C Additional Ablations

C.1 Mask Mode

Table 10: Effect of mask mode on Yelp ($r = 0.3$).

Mask Mode	Attr-Acc ↑	PPL ↓	BLEU ↑	BERTScore ↑
Random	—	—	—	—
Entropy	—	—	—	—
Suffix	—	—	—	—

C.2 Latent Source

C.3 Diffusion Steps

C.4 Temperature

D Efficiency Benchmarks

E LSME Editing Hyperparameters

F Reproducibility

All experiments can be reproduced with the following commands:

```
# LSME editing (Yelp sentiment)
python -m lsme.scripts.run_lsme \
    --checkpoint_path checkpoints/mmdit_latent \
    --latent_dir data/latents \
    --metadata_file data/latents/metadata.json \
    --attribute sentiment --target_value positive \
    --mask_ratio 0.3 --steps 100 \
    --input_file data/yelp_negatives.txt \
    --output_file results/lsme_yelp/output.json

# Evaluate
```

Table 11: Effect of latent source on Yelp ($r = 0.3$, 100 steps).

Latent Source	**Attr-Acc** ↑	**PPL** ↓	**BLEU** ↑	**BERTScore** ↑
No latent (zeros)	—	—	—	—
Random latent	—	—	—	—
Centroid	—	—	—	—
Nearest neighbor	—	—	—	—
Directional ($\alpha = 1.0$)	—	—	—	—

Table 12: Effect of reverse diffusion steps on Yelp ($r = 0.3$, random mask).

Steps	**Attr-Acc** ↑	**PPL** ↓	**BLEU** ↑	**TPS** ↑
10	—	—	—	—
50	—	—	—	—
100	—	—	—	—
500	—	—	—	—

```
python -m lsme.scripts.run_eval \
    --results_file results/lsme_yelp/output.json \
    --output_dir results/lsme_yelp/eval/

# Latent geometry analysis
python -m lsme.scripts.run_geometry \
    --checkpoint_path checkpoints/mmdit_latent \
    --latent_dir data/latents \
    --metadata_file data/latents/metadata.json \
    --attribute sentiment \
    --output_dir results/geometry/
```

G Gap Resolution Summary

Table 13: Effect of sampling temperature on Yelp ($r = 0.3$, 100 steps).

Temperature	**Attr-Acc** ↑	**PPL** ↓	**Distinct-2** ↑	**Self-BLEU** ↓
$T = 0.5$	—	—	—	—
$T = 0.8$	—	—	—	—
$T = 1.0$	—	—	—	—
$T = 1.2$	—	—	—	—

Table 14: Computational cost comparison (Yelp, batch size 32, 512 tokens).

Method	**Params (M)**	**NFE**	**TPS** ↑	**GPU Mem (GB)**
MDLM	—	128	—	—
ReMDM	—	128+	—	—
LatentOps	—	—	—	—
DiffusER	—	—	—	—
LSME (100 steps)	—	100	—	—
LSME (50 steps)	—	50	—	—

Table 15: LSME editing hyperparameters.

Parameter	**Default Value**
Mask ratio	0.3
Reverse diffusion steps	100
Sampling temperature	1.0
Mask mode	random
Latent source	Attribute centroid

Table 16: Gap resolution summary.

Research Gap	**Our Component**	**How It Solves the Gap**
Gap 4: No DLM supports SDEdit-style editing with continuous latent steering	LSME algorithm	Partial mask + latent replacement + joint denoise. Zero architecture changes.
Gap 6: Latent space geometry unexplored in DLMs	SSS & MTS metrics	First formal metrics for DLM latent structure; quantify smoothness and monotonicity.
Gap 7: No comprehensive DLM evaluation suite	DLM-Eval Suite	6-pillar framework covering fluency, control, editing, geometry, diversity, efficiency.